Unit: Unit 0: Foundations of Algebra **Name:** ________________

Topic: GCF LCM **Date:** ____________

Advanced (Habanero)

Q1. Find the GCF of 12 and 18 using prime factorization.
- (A) 2
- (B) 3
- (C) 6
- (D) 9
- (E) 12

Q2. Find the LCM of 8 and 12 using multiples.
- (A) 12
- (B) 16
- (C) 24
- (D) 32
- (E) 48

Q3. What is the GCF of 20 and 30?
- (A) 2
- (B) 5
- (C) 10
- (D) 15
- (E) 20

Q4. Find the LCM of 9 and 15 using prime factorization.
- (A) 15
- (B) 30
- (C) 45
- (D) 60
- (E) 90

Q5. A bookshelf has 12 shelves, and each shelf can hold 8 books. If the bookshelf is currently empty, what is the LCM of the number of shelves and the number of books each shelf can hold?
- (A) 12
- (B) 16
- (C) 24
- (D) 32
- (E) 48

Q6. Find the GCF of 24 and 30 using factors.
- (A) 2
- (B) 3
- (C) 6
- (D) 10
- (E) 12

Q7. A group of friends want to share some candy equally. If they have 18 pieces of candy and there are 6 friends, what is the GCF of the number of pieces of candy and the number of friends?
- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 9

Q8. Find the LCM of 6 and 9 using multiples.
- (A) 12
- (B) 18
- (C) 24
- (D) 30
- (E) 36

Open Ended Questions (FRQ)

Q9. Tom has 12 boxes of pens, with 8 pens in each box. If he wants to put the pens into bags of 4 pens each, what is the LCM of the number of pens in each box and the number of pens in each bag? Show your work and explain your reasoning.

Q10. A bakery is making a special batch of cookies for a holiday sale. They need to package 18 cookies into boxes that hold 3 cookies each. What is the GCF of the number of cookies and the number of cookies each box can hold? Explain your answer and show your work.

Chef's Tips

- Chef's Tips: Make sure to check the work and reasoning for FRQ questions 9 and 10.
- Remind students to use prime factorization or lists of factors to find GCF and LCM.
- Encourage students to read the questions carefully and use the correct method to solve the problem.